Design, Streaming and Scale

Topic 10 of 10

1 questions | skeleton and writing space only

Each question: the prompt, the function or class signature, and ruled lines to write your own solution.

1. Restaurant Waitlist: join / delete / find_first_match

MEDIUM | asked 2026-02-01 | data-structure, oop-design

Onsite coding round: design and implement a restaurant waitlist data structure supporting join, delete, and find_first_match(table_size). The find call returns the earliest-joined user whose party size fits the given table. Implement a class with these methods: join(user) -- append user (with their party_size) to the waitlist. delete(user) -- remove user from the waitlist (could be anywhere in the queue).

```
class Waitlist:

    def __init__(self):
```

```
    def join(self, name, size):
```

```
    def delete(self, name):
```

```
    def front(self, queue):
```

```
    def find_first_match(self, table):
```